NETWORKING

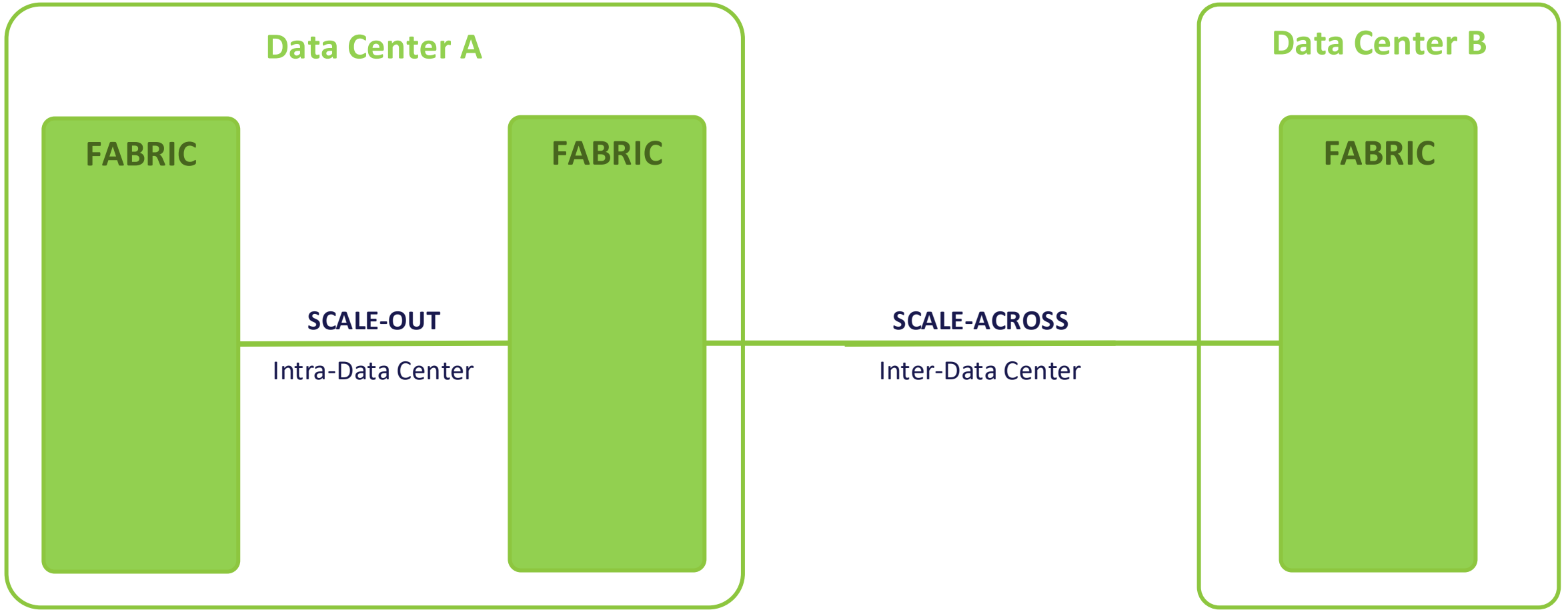
Zero-Touch OCS Fabrics for AI Scale-Out Infrastructure at Hyperscale

Kannan Raj *Architect, Oracle Cloud Infrastructure*

Michael DeMerchant *Product Line Manager, OCS, Lumentum Operations LLC*

OCP BRONZE™ Member

Key Use-Cases for OCS



Three Core Constraints

1. SOFTWARE
Standardize the lifecycle

- One behavior contract
- Zero-touch every task
- Multi-vendor conformance

2. SUPPLY
Scale qualified capacity

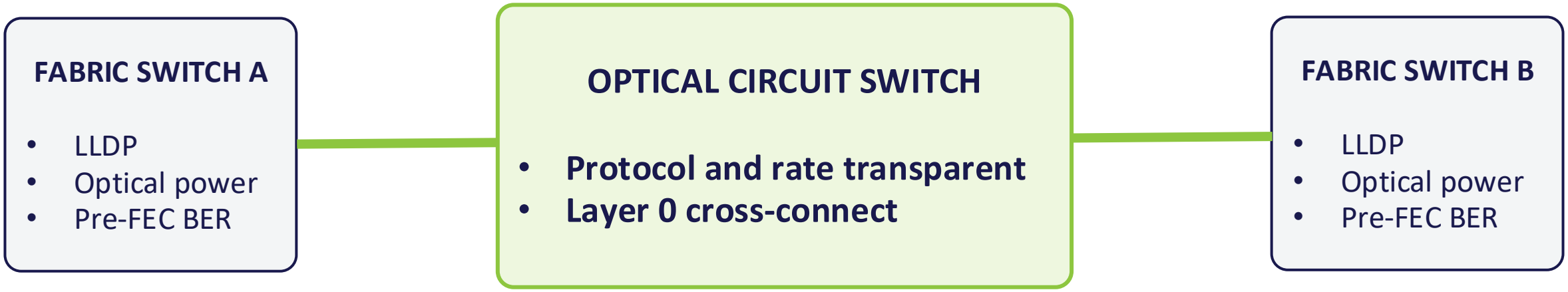
- Reliable availability
- Validated alternates
- Repeatable quality

3. OPTICAL PERFORMANCE
Link budget matters

- Port-to-port consistency
- End-to-end loss budget
- Scale-out + scale-across

OCS fabrics depend on predictable software, supply and optical performance.

OCS changes the operating contract



Packet inspection exists at the endpoints

The OCS does not inspect frames or count packets

Operational state must be inferred end to end.

Zero-touch spans the full lifecycle

DAY 0
Trust the device

- Bootstrap image
- Establish identity
- Enroll credentials

DAY 1
Prove the fabric

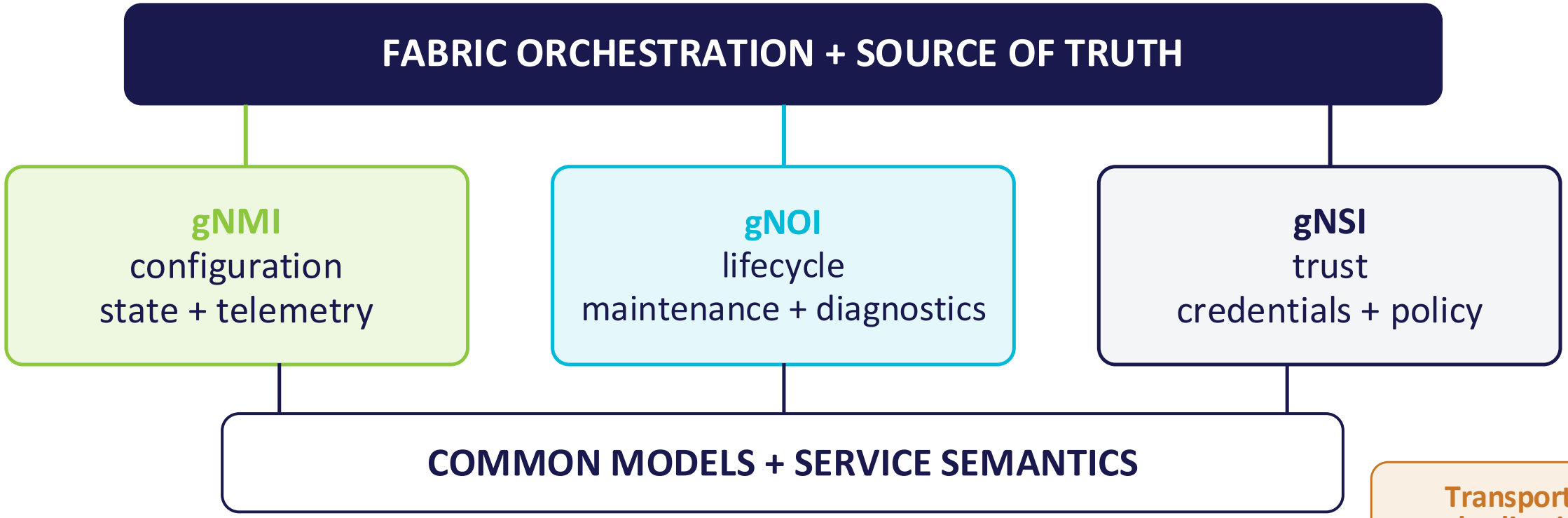
- Apply intent
- Discover topology
- Qualify every path

DAY 2+
Maintain the network

- Stream health
- Upgrade and rotate
- Recover or retire

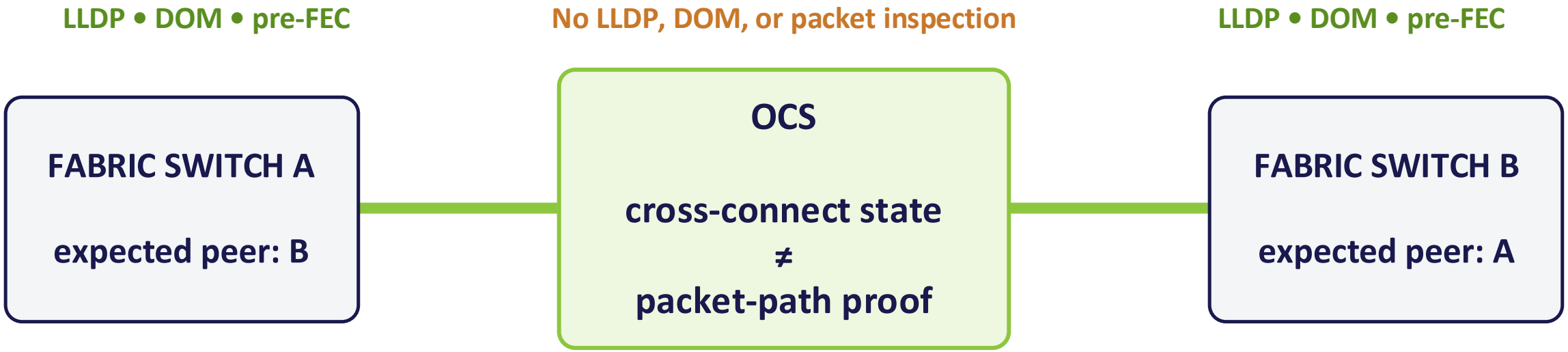
If any phase requires a console, the fabric is not zero-touch.

A common API stack is not enough



Transport standardization is only the start. Behavior and conformance must match too.

OCS creates a validation blind spot



Light can be present on the wrong path.
A green port is not proof of the intended topology.

Patterned OCS pairs make cabling testable

Standard loopback is not enough. Cabling errors require peer-to-peer.

OCS A — PATTERN 1

Spine 01 interface A	→	port 01	→	port 04	→	**Spine 04** interface A
Spine 02 interface A	→	port 02	→	port 03	→	**Spine 03** interface A

OCS B — MIRRORED PATTERN

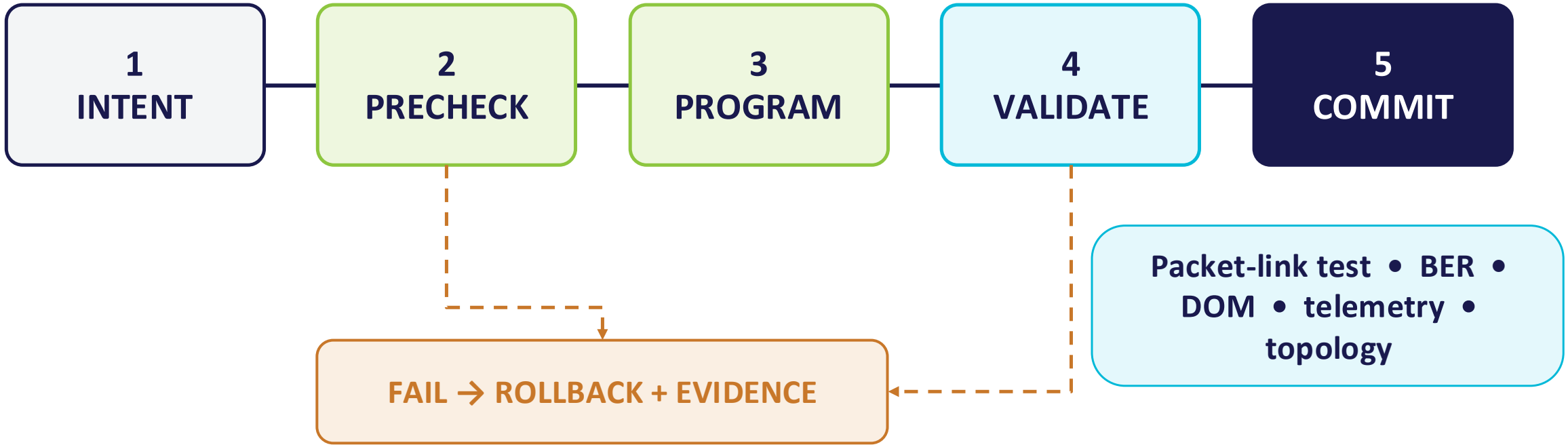
Spine 01 interface B	→	port 01	→	port 03	→	**Spine 03** interface B
Spine 02 interface B	→	port 02	→	port 04	→	**Spine 04** interface B

UNIQUE PEER
Each interface expects a different adjacency

FAILS LOUDLY
Wrong port or wrong OCS changes the observed peer

PARALLEL PROOF
Many links can be qualified without serial loopbacks

Closed-loop automation makes change safe



Every change is a transaction—not a command.

OCS Production At Hyperscale

2. SUPPLY
Scale qualified capacity

- Reliable availability
- Validated alternates
- Repeatable quality

- Deployment at scale requires steady volume supply

The engineering challenge is to:

- Design a product that can be mass produced

AND

- Design the production line that produces it

Our Approach To Scalability

Leverage past learnings



- Telecom experience shaped OCS
- MEMS design and process know how
- Many WSS & Transport line card manufacturing processes can apply

Co-design product, factory and supply chain



- Engage operations from day 1
- Design for manufacturability
- Replicable assembly line
- Multi-source, supply agreements, buffers

Automate Critical Tasks

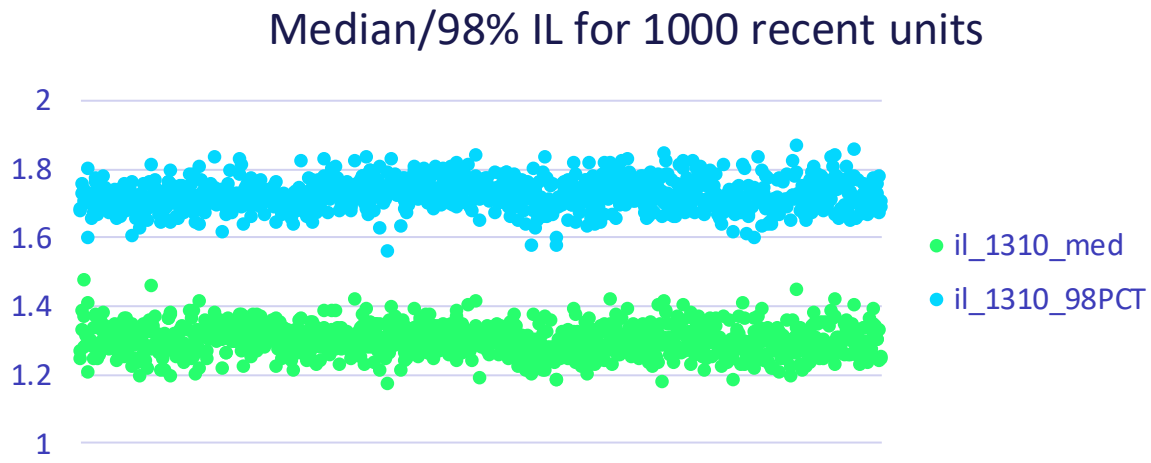


- Optical alignment
- MEMS packaging
- Calibration and Testing
- Visual inspection

Performance And Optical Interface Compatibility

3. OPTICAL PERFORMANCE
Link budget matters

- Port-to-port consistency
- End-to-end loss budget
- Scale-out + scale-across



Standard	IL Budget	Compatibility
IEEE FR/FR4/FR8	4 dB	✓
IEEE LR/LR4/LR8	6.3 dB	✓
OIF ZR/ZR+	11+ dB	✓

OCP needs an operator-ready OCS profile

FOUNDATION	WHAT IT GIVES US	WHAT MUST BECOME NORMATIVE
OpenConfig + gNMI/gNOI/gNSI	Transport, common models, lifecycle and security services	**One mandatory OCS lifecycle profile**
OCP OCS charter	Correct targets: SONiC, models, telemetry, testing, interoperability	**Published behavior, semantics, and artifacts**
Operator requirements	Day 0 / Day 1 / Day 2 evidence and real failure scenarios	**Shared conformance suite across vendors**

Standardized behavior still depends on predictable optical hardware.

Call to Action

- Help define and implement OCS standardization to make it operator-ready
- Contribute your time, experience and expertise

- Get involved with the OCP OCS Sub-Project
  - Biweekly meetings Wednesdays at 9am PST
  - Project Site: https://www.opencompute.org/community/optical-circuit-switching
  - Mailing List: https://ocp-all.groups.io/g/ocp-optical-circuit-switching



- Get involved with the SONiC OCS working group
  - Biweekly meetings Wednesdays at 5pm PST
  - Mailing List: https://lists.sonicfoundation.dev/g/sonic-ocs-wg

Thank You!

Q&A

